AQFP Cryogenic Rack

Design Package

Revision 0.7 — March 2026

Prepared for deployment at Vertiv, CBA, and AM

Executive Summary

This document describes the design of a self-contained cryogenic rack for deploying AQFP (Adiabatic Quantum Flux Parametron) superconducting processors at 4.2 K. The system is built entirely from off-the-shelf and standard vacuum components, housed in a 19-inch equipment rack, and designed for portability between sites.

Key specifications:

PARAMETER	VALUE
Operating temperature	4.2 K (second stage), ~40 K (first stage)
Cryocooler	SHI RDK-101D + CNA-11RC (GM, air-cooled)
Cooling capacity	100 mW @ 4.2 K, ~6 W @ 45 K
Thermal margin	>90% on both stages (with MLI)
Power	120V / 15A standard wall outlet (~2 kW total)
Cooling water	Not required
Form factor	19-inch rack + floor-standing compressor
Vacuum	HiCube 80 Eco turbo station, $<10^{-8}$ Torr ultimate
Magnetic shielding	FINEMET (40 K) + superconducting Pb foil (4 K)
Cooldown time	~60–90 minutes to 4 K

Estimated cost: ~$52k for the rack build, plus ~$11k for external test equipment (Yokogawa GS200 + Keithley 2182A) if not already available.

Timeline: 4 weeks from procurement start to operational, gated by the 12-week cryocooler lead time.

Deployment plan: Vertiv (first), then CBA and AM. The system is designed to be relocated with minimal effort — standard power, air-cooled, modular construction.

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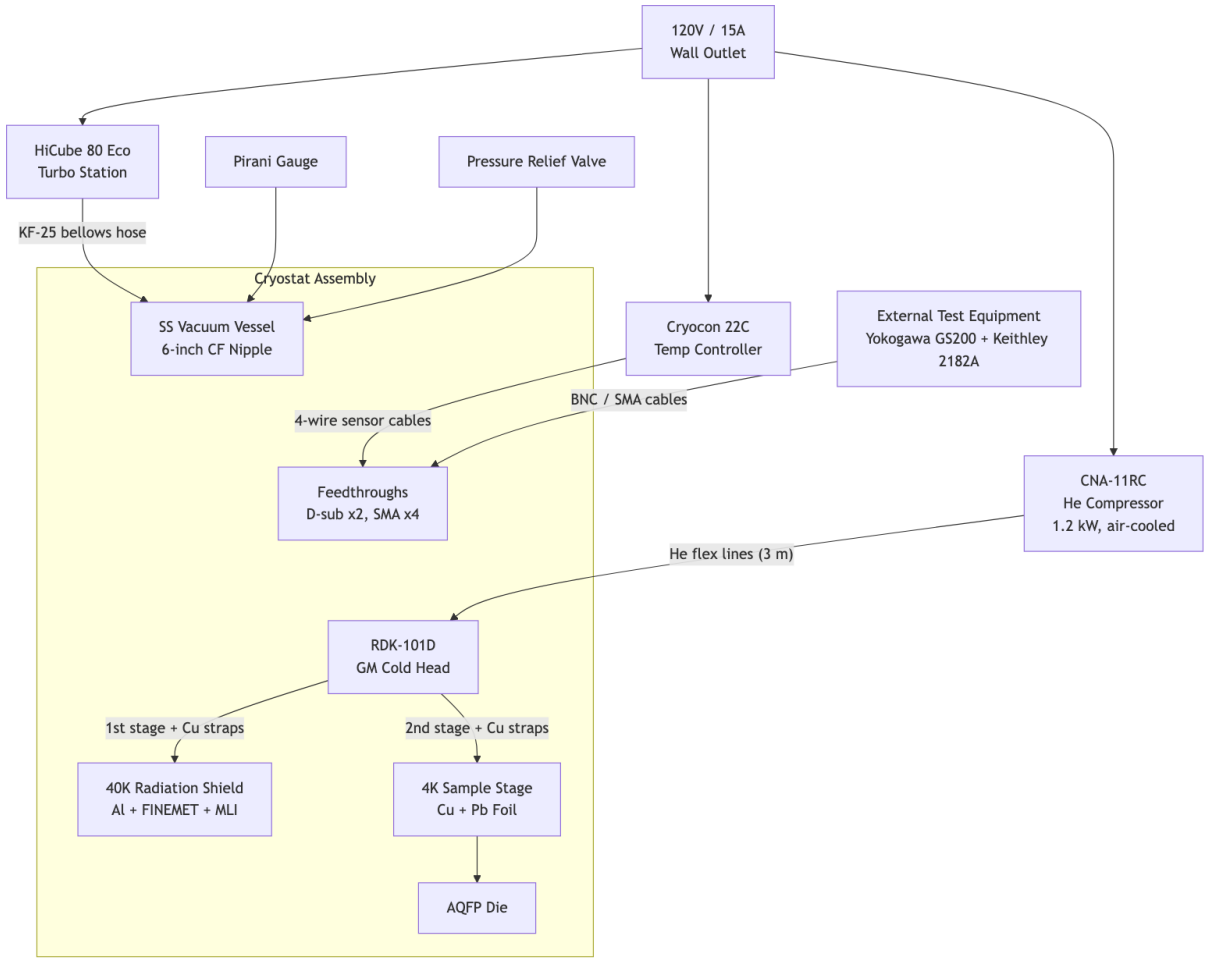
1. System Architecture

The cryo rack is a self-contained cryogenic deployment platform. A GM cryocooler, vacuum system, magnetic shielding, and instrumentation are housed in a standard 19-inch rack, with all signal and power connections accessible from the front or rear panel.



Thermal stages

The system has three thermal stages:

- **300 K (room temperature):** Vacuum vessel walls, feedthroughs, pump, compressor, and all room-temperature electronics.
- **~40 K (first stage):** Aluminum radiation shield with FINEMET nanocrystalline magnetic shielding and MLI blankets. Wiring is thermally intercepted here.
- **4.2 K (second stage):** OFHC copper sample stage where the AQFP die is mounted, surrounded by superconducting lead foil for residual magnetic field attenuation.

Signal path

DC bias currents are delivered via phosphor bronze twisted pairs, heat-sunk at 40 K and 4 K. AC clock signals travel on semi-rigid stainless steel coaxial cables, also thermally intercepted at both stages. All connections terminate at room-temperature feedthroughs on the cryostat top plate.

Ground isolation

The cold head body is galvanically continuous with the compressor chassis through the stainless steel flex lines. The compressor is earth-grounded through its AC power cord. The GM cycle valve motor produces periodic current transients (~1 Hz) that propagate through this ground structure.

The system uses two isolated ground domains (see ADR-004):

- **Domain 1 (earth ground):** Building mains → compressor chassis → flex lines → cold head → vacuum jacket → rack chassis. Carries compressor motor currents. No measurement signals.
- **Domain 2 (isolated measurement ground):** Isolation transformer secondary → readout electronics → feedthrough pins → copper backplane at 4 K → AQFP circuit ground. Signal and bias currents only.

Isolation is maintained by:

LOCATION	COMPONENT	FUNCTION
4 K (cold finger)	Polycrystalline alumina disk, 25–30 mm, indium foil both faces	Breaks galvanic path between cryocooler and circuit ground
300 K (feedthroughs)	Ceramic-to-metal hermetic seals	Pins isolated from flange body by design
300 K (power entry)	Tripp Lite IS500HG isolation transformer (500 W, Faraday-shielded)	Isolates readout ground from building mains

Critical assembly discipline: No readout cable shield, coax braid, ground lug, or debug probe ground may touch the cryostat body, vacuum jacket, or rack chassis during measurement.

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2. Cryostat Assembly

Design philosophy

All vacuum hardware is standard CF/KF catalog items sourced from multiple vendors (Kurt Lesker, MDC, Nor-Cal, Ideal Vacuum). Only the radiation shield and sample stage are custom machined — simple aluminum and copper parts any machine shop can make. No single-vendor lock-in.

The cryostat is top-loading: the cold head hangs from the top plate with the cold end pointing down. Feedthroughs are on the top plate or welded side ports. Pumping connects via KF-25 ports on the vessel body.

Cross-section

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┌──────────────────────────────┐
│         COMPRESSOR           │  CNA-11RC, 120V, on floor / rack shelf
│         (300 K)              │
└──────────────────────────────┘
            │ flex lines (He supply + return)
            │ cold head cable (motor drive)
            │
═══════════╪═══════════════════════════════  ← TOP PLATE (300K)
│           │                FT  FT  FT  │    8" CF multiport cluster flange
│     ┌─────┴─────┐          ○   ○   ○   │    FT = feedthrough ports (D-sub, SMA)
│     │ COLD HEAD │                      │    Center bore: CF seal to cold head
│     │  (motor)  │                      │    Viton O-ring seal to vessel body
│     │           │                      │    (reusable for sample access cycling)
│     │           │                      │
│     ├───────────┤ ← 1st stage          │
│     │  (40 K)   │   flange             │
│   ┌─┴───────────┴─┐                    │
│   │               │                    │
│   │  strap│  strap│                    │  ← OFHC Cu thermal straps (x2)
│   │       │       │                    │
│   │   ┌───┴───┐   │                    │
│   ├───┤ RAD   ├───┤  ·····             │  ← RADIATION SHIELD (40K)
│   │   │ SHIELD│   │  :MLI:             │    6061 Al cylinder + end caps
│   │   │ (Al)  │   │  :   :             │    FINEMET ribbon wrapped on surface
│   │   │       │   │  :   :             │    MLI blankets on outside of shield
│   │   │FINEMET│   │  :   :             │    Wires heat-sunk here (GE varnish)
│   │   │wrapped│   │  :   :             │
│   │   │       │   │  :   :             │
│   │   │  ┌────┐   │  :   :             │
│   │   │  │2nd │   │  :   :             │  ← 2nd STAGE (4K) cold finger
│   │   │  │stg │   │  :   :             │
│   │   │  ├────┤   │  :   :             │    Cu strap from 2nd stage
│   │   │  │    │   │  :   :             │
│   │   │  │ ┌──┐   │  :   :             │  ← SAMPLE STAGE (4K)
│   │   │  │ │P │   │  :   :             │    OFHC Cu plate, gold-plated
│   │   │  │ │b │   │  :   :             │    Pb foil wrapped around DUT area
│   │   │  │ ├──┤   │  :   :             │    Cernox sensor mounted here
│   │   │  │ │D │   │  :   :             │  ← DUT PCB CARRIER
│   │   │  │ │U │   │  :   :             │    Wirebonded AQFP die
│   │   │  │ │T │   │  :   :             │    SMA connectors for RF
│   │   │  │ └──┘   │  :   :             │
│   │   │  └────┘   │  :   :             │
│   │   └───────────┘  :···:             │
│   └─────────────────┘                  │
│                                        │  ← VACUUM VESSEL BODY (300K)
│  ○ KF-25 (pump)                        │    SS CF nipple, 6" OD tube
│  ○ KF-25 (gauge)                       │    KF stubs welded on sides
│  ○ KF-25 (relief valve)                │
│                                        │
═══════════╪═══════════════════════════════  ← BOTTOM PLATE (300K)
           │                               Blank 8" CF flange
           │            │
           │            │
┌──────────┴──────┐  ┌──┴──────────┐
│ HiCube          │  │ Pirani      │
│ 80 Eco          │  │ gauge       │
│ (pump)          │  │ CVM211      │
└─────────────────┘  └─────────────┘
```

Assembly layers

300 K — Room temperature

#	PART	MATERIAL	INTERFACE
1	Equipment rack	Steel	Floor
2	Compressor (CNA-11RC)	—	120V wall outlet
3	Flex lines + cold head cable	SS/Cu	Compressor to cold head
4	Vacuum vessel body	304L SS	8” CF top + bottom
5	Top plate	304L SS	8” CF, center bore for cold head
6	Bottom plate	304L SS	8” CF blank
7	Viton O-ring (top plate seal)	Viton	Reusable sample access joint
8	CF copper gaskets	OFHC Cu	Permanent joints
9	KF side ports	SS	Welded on vessel body
10	Pressure relief valve	—	KF-25 port on vessel
11	Vacuum gauge (Pirani)	—	KF-25 port
12	HiCube turbo pump station	—	KF-25 bellows hose to vessel
13	D-sub feedthroughs (x2)	—	CF ports on top plate
14	SMA feedthroughs (x4)	—	CF ports on top plate

40 K — First stage

#	PART	MATERIAL	INTERFACE
15	Cold head (RDK-101D)	—	CF seal through top plate
16	Thermal straps (x2)	OFHC Cu braid	1st stage flange to rad shield
17	Indium wire	In	Crushed at bolted joints
18	Radiation shield	6061-T6 Al	Cylinder + end caps
19	FINEMET ribbon	Nanocrystalline	Wrapped on rad shield surface
20	MLI blankets	Al mylar + dacron	Wrapped around shield exterior
21	Si diode sensor	—	Mounted on shield
22	Wire heat sinking	GE varnish	Wires anchored to rad shield

4 K — Second stage

#	PART	MATERIAL	INTERFACE
23	Thermal straps (x2)	OFHC Cu braid	2nd stage to sample stage
23b	Alumina isolation disk	Al_2O_3 (polycrystalline)	Between cold finger and sample stage (see §1, ground isolation)
24	Sample stage	OFHC Cu	Clamped to cold finger through alumina disk
25	Lead foil shield	Pb, 0.1 mm	Wrapped around DUT area
26	Cernox sensors (x2)	—	2nd stage + DUT mount
27	DUT PCB carrier	FR4/Cu	Bolted to sample stage

Wiring path

```
External instruments (GS200, 2182A)
        │ DC: BNC/banana cables
        │ RF: SMA cables
        ▼
   ┌──────────────┐
   │ Feedthroughs │  D-sub (DC), SMA (RF) — hermetic, on top plate
   └──────────────┘
          │
          │ PhBr wire (DC), SS coax (RF) — inside vacuum
          │
      ┌───┴────┐
      │ 40K    │  Wires thermally anchored (GE varnish on bobbins)
      │ shield │  Intercepted heat goes to 1st stage
      └────────┘
          │
          │
      ┌────────┐
      │ 4K DUT │  DC bias pads + SMA edge-launch on PCB
      │ carrier│  Wirebonded to AQFP die
      └────────┘
```

Vacuum path

```
  Atmosphere
       │
       │
   ┌───┴────┐
   │ HiCube │  Turbo + diaphragm backing pump
   │ 80 Eco │  KF-25 bellows hose to vessel
   └────────┘
       │
       │
   ┌───┴────┐
   │ Vessel │  SS CF nipple body
   │ interior│  Pirani gauge on KF-25 side port
   │        │  Relief valve on KF-25 side port
   └────────┘
       ↓ at 4K, cryopumping maintains <1e-6 Torr
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Modularity and cryocooler swapping

The top plate is the only cryocooler-specific part. If the cold head changes (e.g., from RDK-101D to BMC401 or a pulse tube), only the top plate needs replacement or re-machining (~$400 cost). All other cryostat components are cold-head agnostic.

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3. Thermal Budget

Cryocooler capacity

RDK-101D with CNA-11RC compressor at 60 Hz (US power):

STAGE	TEMPERATURE	CAPACITY
1st stage	45–60 K	3–5 W
2nd stage	4.2 K	**100 mW**

4 K (second stage) heat budget

SOURCE	LOAD (MW)
Radiation (40 K to 4 K)	0.3
DC wire conduction (16x PhBr 36AWG)	0.1
RF coax conduction (4x SS 0.086”)	2.0
DUT dissipation (AQFP: nW–uW)	~0
Total	**~2.4**
Available	**100**
Utilization	**2.4%**

40 K (first stage) heat budget

SOURCE	WITH MLI	WITHOUT MLI
Radiation (300 K to 40 K)	120 mW	1,800 mW
DC wire conduction	3 mW	3 mW
RF coax conduction	35 mW	35 mW
Total	**~160 mW**	**~1,840 mW**
Available	**~3,000 mW**	**~3,000 mW**
Utilization	**5%**	**61%**

Both stages have large thermal margin. MLI is recommended but not strictly required at the baseline shield size (100 mm diameter). For larger shields (200–300 mm), MLI is mandatory — see appendix.

Shield sizing (how big can we go?)

With MLI, shield size is thermally unconstrained for any rack-mount design:

SHIELD (DIA X H)	AREA	WITH MLI	WITHOUT MLI
100 x 150 mm	0.063 m²	95 mW **(3%)**	1.4 W (48%)
200 x 300 mm	0.251 m²	376 mW **(13%)**	5.8 W (OVER)
300 x 400 mm	0.518 m²	777 mW **(26%)**	12 W (OVER)

A 200–300 mm diameter shield provides generous working space for multiple DUTs and complex wiring, uses only 13–26% of the first-stage budget with MLI, and fits comfortably in a 19-inch rack (450 mm between rails).

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4. Cryocooler Selection

Comparison of quoted options

PARAMETER	SHI RDK-101D	SHI RDC-02K	BAMA BMC401	CRYOMECH PT205
Type	GM	GM	GM	Pulse tube
Quoted price	$27,230	$37,155	$13,000	$42,325
Lead time	12 weeks	12 weeks	4 weeks	36–40 weeks
4 K capacity	0.1 W	0.2 W	0.16 W	0.1–0.3 W
40 K capacity	~6 W @ 45 K	~6 W @ 45 K	3 W @ 45 K	TBD
Min temp	~3.5 K	< 2 K	< 2.3 K	5–6 K
120V / 15A	**Yes**	**Yes**	**No** (220V)	**Likely yes**
Cooling	Air	Air	Unconfirmed	Air
Compressor power	~1.2 kW	~1.2 kW	~1.5 kW	~1.5 kW
Maintenance	10,000 hr	10,000 hr	10,000 hr	~20,000 hr
Vibration	Moderate (GM)	Moderate (GM)	Moderate (GM)	Low (PT)

Selection rationale

Primary: SHI RDK-101D + CNA-11RC — 120V plug-and-play, air-cooled, well-established platform with extensive support. 12-week lead time.

Upgrade option: SHI RDC-02K + CNA-11RC — same compressor and rack integration as the RDK-101D, but 2x the 4 K capacity (0.2 W) and guaranteed < 2 K base temperature. $10k premium over the RDK-101D. Relevant if Phase 2 wiring loads grow or sub-4 K operation is needed.

Backup: Bama BMC401 — half the cost, 4-week delivery, but 220V/50Hz compressor is not compatible with standard 120V/15A outlets. Under evaluation for 60 Hz compatibility and potential pairing with SHI CNA-11RC compressor.

Not selected: Cryomech PT205 — 36–40 week prototype lead time and highest cost eliminate it for near-term build. Low vibration and longer maintenance interval do not materially benefit AQFP electrical testing. Reconsidering for future permanent installations.

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5. Bill of Materials

Cost summary

ASSEMBLY	ITEMS	EST. COST
cryocooler-system	1	$27,230
cryostat-vessel	5	$3,530
vacuum-plumbing	3	$9,300
magnetic-shielding	2	$830
thermal-wiring	5	$750
temperature-monitoring	3	$2,650
test-equipment	2	$10,700
rack-utilities	9	$3,980
Total		**$58,970**

cryocooler-system

PART	VENDOR	QTY	UNIT COST	LEAD (WK)	STATUS
Option A: SHI RDK-101D Complete System	Sumitomo / SHI Cryogenics of America	1	$27,230	12	quoted

cryostat-vessel

PART	VENDOR	QTY	UNIT COST	LEAD (WK)	STATUS
Vacuum Vessel Body (CF Nipple or 6-Way Cross)	Kurt Lesker / MDC / Nor-Cal	1	$1,500	1	planned
Top Plate (Multiport Cluster Flange or Bored Blank)	Kurt Lesker / MDC	1	$800	2	planned
Radiation Shield (40K) — Custom Machined Aluminum	Local machine shop / Protolabs / Xometry	1	$800	2	planned
Sample Stage (4K) — Custom Machined Copper	Local machine shop	1	$400	2	planned
Alumina Electrical Isolation Disk (4K)	CoorsTek / McMaster-Carr	1	$30	1	planned

vacuum-plumbing

PART	VENDOR	QTY	UNIT COST	LEAD (WK)	STATUS
Vacuum Pumping: Turbo Station (Pfeiffer HiCube 80 Eco)	Pfeiffer Vacuum	1	$7,900	1	planned
Vacuum Gauge (Convection Pirani)	InstruTech	1	$400	1	planned
Electrical Feedthroughs (DC, Multi-pin)	MDC Vacuum / Kurt Lesker	2	$500	2	planned

magnetic-shielding

PART	VENDOR	QTY	UNIT COST	LEAD (WK)	STATUS
FINEMET Nanocrystalline Ribbon	Proterial (formerly Hitachi Metals)	1	$800	2	planned
Lead Foil (Superconducting Shield)	McMaster-Carr / Alfa Aesar	1	$30	1	planned

thermal-wiring

PART	VENDOR	QTY	UNIT COST	LEAD (WK)	STATUS
Copper Braid / Thermal Straps (40 K stage)	McMaster-Carr	2	$50	1	planned
Copper Braid / Thermal Straps (4 K stage)	McMaster-Carr	2	$50	1	planned
Phosphor Bronze Wiring Loom (DC)	California Fine Wire / MWS Wire	1	$50	1	planned
PCB DUT Mount / Chip Carrier	Custom / PCBWay	2	$200	2	planned
MLI Blankets (Multi-Layer Insulation)	DIY or Dunmore / FrakoTerm	1	$100	1	planned

temperature-monitoring

PART	VENDOR	QTY	UNIT COST	LEAD (WK)	STATUS
Silicon Diode Temperature Sensor (4 K)	Lake Shore Cryotronics	2	$50	1	planned
Silicon Diode Temperature Sensor (40 K)	Lake Shore Cryotronics	1	$50	1	planned
Temperature Controller	Cryocon (Cryogenic Control Systems)	1	$2,500	2	planned

test-equipment

PART	VENDOR	QTY	UNIT COST	LEAD (WK)	STATUS
DC Voltage/Current Source	Yokogawa	1	$4,000	2	planned
Nanovoltmeter	Keithley / Tektronix	1	$6,700	2	planned

rack-utilities

PART	VENDOR	QTY	UNIT COST	LEAD (WK)	STATUS
Equipment Rack (19” Standard)	Vertiv / APC / Chatsworth	1	$1,500	1	planned
Rack PDU (Metered)	Vertiv / APC	1	$400	1	planned
Isolation Transformer (Readout Electronics)	Tripp Lite	1	$250	1	planned
KF/CF Vacuum Fittings and Hardware	Kurt Lesker / MDC	1	$800	1	planned
Pressure Relief Valve (Burst Disc)	Kurt Lesker / MDC	1	$150	1	planned
CF Copper Gaskets (Consumable)	Kurt Lesker / MDC	10	$20	1	planned
Indium Wire (Thermal Interface)	Indium Corporation / Alfa Aesar	1	$80	1	planned
Wire Heat Sinking Supplies	Lake Shore / CMR Direct	1	$100	1	planned
Miscellaneous Hardware	McMaster-Carr / Thorlabs	1	$500	1	planned

Full BOM is auto-generated from `bom/bom.yaml` (rev 0.6) by the build script. The tables below show active items only (qty > 0). Reference-only options (alternative cryocoolers, scroll pump) are excluded from cost totals.

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6. Site Requirements — Vertiv

Power

REQUIREMENT	SPECIFICATION
Outlet type	NEMA 5-15 (standard US 120V / 15A)
Circuits needed	2 recommended (compressor on one, instruments on another)
Total power draw	~2–3 kW
Compressor	~1.2 kW (CNA-11RC), direct to building mains (earth-grounded)
Turbo pump	~150 W (HiCube 80 Eco)
Instruments	~200 W behind isolation transformer (Tripp Lite IS500HG, 500 W)

No special electrical work required. Standard office/lab outlets are sufficient. The readout electronics are powered through a rack-mounted isolation transformer to maintain a clean, isolated measurement ground (see §1, ground isolation).

Space

REQUIREMENT	SPECIFICATION
Rack	19-inch standard, ~20U minimum
Rack footprint	~0.6 x 0.8 m
Compressor	Floor or shelf adjacent to rack, ~0.5 x 0.5 m
Clearance	Allow ~0.5 m behind rack for cable routing and airflow
Cryostat position	Top of rack or adjacent bench

Environment

REQUIREMENT	SPECIFICATION
Ambient temperature	15–30 C (standard lab/office)
Cooling water	**Not required** (air-cooled compressor)
Ventilation	Standard room HVAC sufficient; compressor rejects ~1–2 kW as heat
Floor loading	~150–200 kg total (rack + compressor + cryostat)
Vibration isolation	Not required for AQFP testing

Network (optional)

- Ethernet for Cryocon 22C remote temperature monitoring
- USB for instrument control (GS200, 2182A)

Facility preparation checklist

- [ ] Confirm availability of 2x 120V / 15A outlets near rack location
- [ ] Confirm 19-inch rack (Vertiv may supply) or designate rack location
- [ ] Confirm floor space for compressor adjacent to rack
- [ ] Confirm adequate room ventilation for ~2 kW heat rejection
- [ ] Assign Ethernet drop if remote monitoring desired

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7. Timeline

Target: Operational cryo rack within 4 weeks of procurement start.

WEEK	ACTIVITIES
1	Finalize cryocooler selection. Send RFQs. Order long-lead items. Complete cryostat detailed design.
2	Place all orders. Commission custom machining (radiation shield, sample stage). Begin rack layout.
3	Receive and inspect components. Assemble cryostat. Install magnetic shielding. Wire feedthroughs. Room-temperature functional tests.
4	Leak check vacuum system. Pump down and cooldown. Verify base temperature. Measure residual magnetic field. Functional test with AQFP die (if available).

Critical path

The 12-week cryocooler lead time (SHI RDK-101D) gates the overall schedule. All other components have 1–2 week lead times. During the cryocooler wait:

- Complete cryostat detailed design using outline drawing dimensions
- Procure and assemble all non-cryocooler components
- Perform room-temperature vacuum and wiring tests
- Prepare DUT carriers and test fixtures

If BMC401 (4-week lead) is selected as backup, the full build can be completed in approximately 5–6 weeks end-to-end.

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8. Open Items and Risks

Open actions

#	ACTION	OWNER	STATUS
1	Re-quote SHI RDK-101D (Quote 5618 expired). Specify CF flange option.	Procurement	Pending
2	Confirm Cryomech PT205 120V availability when re-quoting	Procurement	Pending
3	Clarify Bama BMC401 mounting pattern, 60 Hz compatibility, SHI compressor compatibility	Engineering	Pending
4	Finalize cryostat vessel length from RDK-101D outline drawing dimensions	Engineering	Pending
5	Detailed top plate port layout (feedthrough count and placement)	Engineering	Pending
6	DUT PCB carrier design (depends on AQFP die pinout)	Engineering	Pending

Risk register

RISK	IMPACT	MITIGATION
Cryocooler lead time (12 wk)	Schedule slip	BMC401 backup (4 wk). Pre-assemble all other components.
RDK-101D quote expired	Price/availability change	Re-quote immediately. Budget has margin.
BMC401 not 120V compatible	Limits portability	Investigate SHI CNA-11RC + BMC401 cold head hybrid.
Custom machining delays	1–2 week slip	Use Protolabs/Xometry for fast-turn; simple parts.
MLI installation quality	Higher thermal load	>90% margin on both stages absorbs poor MLI.
Vacuum leak at O-ring seal	Pumpdown failure	Turbo pump can maintain vacuum continuously. Viton adequate to 10^{-7} Torr.

Appendix A: MLI Sourcing

10-layer MLI blankets are recommended for radiation shields larger than 150 mm diameter. Material needed for a 200 mm x 300 mm shield: ~2 m² each of reflector and spacer.

OPTION	MATERIAL	COST	LEAD TIME
DIY (recommended)	Aluminized mylar (Amazon) + polyester netting (Joann Fabric)	$50–75	Days
Professional	Double-aluminized PET (Dunmore) + Dacron netting	$300–500	1–2 weeks
Pre-made blankets	Custom blankets (FrakoTerm, Poland)	$500+	2–4 weeks

DIY MLI has been validated by WSU HYPER lab as comparable in thermal performance to professional MLI. With >90% thermal margin, DIY is more than adequate.

MLI adds ~15–30 minutes to pumpdown time due to additional outgassing surface area. This is a manageable trade-off.

Appendix B: 3D Printed Parts (PLA at 4 K)

PLA has been validated at 4 K for non-thermally-conductive structural parts. Low thermal conductivity is a feature for thermal isolation.

Suitable for 3D printing:

PART	BENEFIT
Wire routing bobbins (40 K)	Custom geometry for wire routing
DUT mount adapter	Fast iteration as DUT pinout changes
Magnetic shield brackets	Hold FINEMET + Pb foil in position
Thermal isolation standoffs	Low-conductivity spacers between stages
Feedthrough strain reliefs	Cable management

Must remain metal: Radiation shield (Al), sample stage (Cu), vacuum vessel (SS), thermal straps (OFHC Cu).

PLA outgasses at room temperature in vacuum but this is handled by the turbo pump during pumpdown (adds a few minutes). Once cold, outgassing drops to near zero.